Project Description

For the second LSM project, we are looking at using real world data or more precisely real outer space data form the Gaia RD3 [1] dataset of stars. This is a large dataset of 626,000 stars spread out on 6 different types of stars. The idea is to simulate gravity between the stars with a gridless direct-summation N-body approach. This requires communication of the stars relative positions and their gravitational effect upon each other for each iteration.

Initial Implementation plan

Using a much smaller subset (1000 of each of the star types) we use the "Allgather" method so every rank will hold and receive all star positions. This should allow us to test our setup and our timesteps.

1. Each rank owns N/P stars
2. Compute acceleration of each star by summing all other stars gravitational force on that star.
3. Calculate new star velocities and update positions
4. Validate with energy conservation

Once we have this setup we can create several baselines. Each with varying quantities of stars. Next step would be to incorporate varying mass estimations. After which we can begin testing different communication methods.
The data provided will give us the stars 3D location relative to the Sun, each star will also have a velocity, mass estimate and its id. This will require a unique data structure. In later iterations we may add more detail in the struct or other data structs like a star cluster.
As we begin scaling the calculations the "Allgather" method will most likely reach a communication floor forcing us to look at communication methods and tricks to continue to run simulations within reasonable computational time. Initially, this project will have equal load balance equal number of stars per rank. However, if we add a gravitational affect threshold (a radius of the stars gravitational effective influence) then ranks will have different calculation sizes. As we saw in project 1, dynamic scheduling was very useful in reducing waste time when there is load imbalance without much overhead added. So we will implement that method here.
Our overall goal is to simulate with as many of the stars as possible, however, as we do not want to sit waiting for extremely long wait periods we will look at maximum simulation times of 15 minutes (we will define this specifically after the first subset runs). Our goal will be As we expect the speed up to not follow Amhdal's law, we do not expect the doubling of the amount of cores to halve the calculation time. We will therefore look at when this doubling stops having a significant effect. Using that number of cores, we will then start to look at grouping strategies for the stars. We want to look at communication in collectives and in groups to see how many more stars we can simulate with the different adjustments, methods and parameter changes.

Validation

For validation we will first run the code with a serial setup, if it passes the 3 tests, we will then run it parallel and compare the results.

- A 2 body problem with the same run time. As this can be calculated numerically so we can then compare the simulation to the actual answer.
- Using the Plummer sphere as a way of verifying the density
- Energy and momentum conservation

Once we have a validated model we can begin with parallelization improvements.

Introduction

Implementation and methodology

Experiments

Overall Conclusion

AI declaration

We used an AI assistant (Claude, Anthropic, OpenAI) during the project. Its use differed between the parts of the work:

- **Programming:** Used for writing code, we focused on conducting experiments and understanding the HPC.
- **Understanding and debugging:** explanations of MPI concepts (blocking and non-blocking communication, request completion, eager and rendezvous protocols), help with errors in our send/receive code and with the MPI setup on our laptops.
- **Report:** AI was used for some LaTeX-formatting (tables, figures, layout).
- **Language:** we used the AI to turn our results and notes into clear, correct English, i.e., for help with some of the phrasing, grammar and wording. The content, the numbers and the interpretations come from our own experiments and were checked by us.

References

[1] ESA, *Gaia 626k stars from dr3 (data release 3) v2*. https://www.kaggle.com/datasets/realkiller69/gaia-stars-dataset-from-dr3-data-release-3, 2023. Accessed: 2026-10-27.